RAKESH CHOWDARY POLINA

rpolina@okstate.edu | https://www.linkedin.com/in/rakesh-chowdary-polina | 405-894-3989

EDUCATION

Master of Science in Computer Science **May 2024**
Oklahoma State University, Stillwater, OK **CGPA: 3.909/4**

SKILLS

Languages:	Python, Java, JavaScript, SQL, C, C++, HTML, CSS, Swift
Frameworks:	Flask, Django, React JS, Angular JS, Node JS, FastAPI, React Native
Databases:	MySQL, MongoDB, NoSQL, PostgreSQL, Cassandra
Web Technologies:	RESTful API, Django ORM, SQLAlchemy, OAuth 2, jQuery, JMS, RabbitMQ
Web Services:	REST, SOAP
Tools/Platforms:	Docker, Git, JIRA, Jenkins, Postman, Ansible, Kubernetes, Apache Kafka, Terraform, Swagger
AWS Cloud Services:	EC2, S3, EKS, RDS, ELB, Cloud Watch, CloudFront
Build Tools:	Pip, Poetry

WORK EXPERIENCE

Research Data Analyst, Oklahoma State University **May 2023 – May 2024**

- Engineered and implemented a robust facial recognition pipeline, performing both quantitative and qualitative analysis on image data with Python and InsightFace, to achieve a precision rate of 93.6%
- Implemented DBSCAN clustering and rigorously verified data accuracy across 182,000+ images, enhancing archive reliability and insight precision.
- Designed an intuitive UI for the labeling tool using Flask and React.js, streamlining user engagement and annotations.
- Designed and maintained database structures using SQL, enhancing user interaction management and feedback data integrity, ensuring operational excellence.

Software Engineer, Ericsson, India **Jul 2019 – Jul 2022**

- Transitioned monolithic architecture to microservices with RESTful APIs using Python, Flask, and FastAPI. Integrated Apache Kafka for asynchronous communication, improving system scalability by handling 50% more concurrent users.
- Assisted in website design and development, incorporating user interface enhancements for dashboards and portals. Developed React based front-end solutions with TypeScript for dynamic content generation.
- Performed comprehensive requirement analysis and crafted technical specifications for API development. Reduced system downtime by 30% by implementing robust security mechanisms, optimizing PostgreSQL-based data access layers, and enhancing backend performance using Django. Utilized SQL for complex query design and optimization.
- Designed and implemented scalable APIs adhering to industry standards. Reduced API response time by 40% through asynchronous programming patterns and non-blocking features. Enhanced user experience by integrating real-time data updates and optimizing performance.
- Implemented containerization and orchestration using Docker and Kubernetes. Reduced deployment time by 50% and ensured system resilience in production environments by leveraging AWS services such as ECS and Fargate.
- Streamlined CI/CD pipelines using Jenkins and GitLab, automating testing and deployments. Reduced time-to-market by 40% and minimized production incidents. Integrated AWS services like CodePipeline and CodeBuild to improve pipeline efficiency, incorporating automated unit testing and documentation generation tools.
- Integrated GenAI frameworks into web-based platforms, enabling intelligent user interaction and predictive analytics. Deployed machine learning models using AWS SageMaker and APIs powered by TensorFlow and Hugging Face. Monitored Linux system logs for performance tuning during deployments.
- Adhered to computer security best practices, enforcing secure authentication, data encryption, and access control policies. Ensured compliance with organizational standards and AWS security protocols.
- Followed AGILE (SCRUM) methodologies with bi-weekly sprint planning and daily status meetings to monitor project progress. Collaborated with cross-functional teams, fostering seamless communication and coordination.
- Wrote and reviewed clear technical documentation, ensuring accessibility for stakeholders. Monitored system metrics to address performance concerns and implemented corrective measures effectively using tools like Grafana.

PROJECTS

Sleep data analysis - Applied LSTM network and Granger causality techniques to investigate sleep data on a dataset comprising over 5,000 individuals' sleep patterns, unveiling insights into temporal dynamics and causal connections within the dataset

HireOSUgrads App - Developed an iOS application for university students and recruiters, featuring intuitive swipe gestures for seamless interactions. Utilized Swift, followed MVC architecture and integrated SQLite for robust data management

Journal App - Developed an android application, that allows users to store their daily events and memories using React Native, Express JS, Redux, Redis, MongoDB, REST APIs, Git.